$$\int \frac{\ln(\ln x)}{x}\,dx$$

$u = \ln x$

$du = \frac{1}{x}\,dx$

$$\int \ln(u)\,du$$

$e^x, \sin x, \cos x$

$$\int x^{\wedge}(\cdots)\,dx$$

$$\int (\cdots)\,x^{\wedge}\,dx$$

$\ln x, \sin^{-1}x, \cos^{-1}x, \ldots$

$v = \ln u \quad dv = du$

$dv = \frac{1}{u}du \quad v = u$

$$u\ln u - \int u \cdot \frac{1}{u}\,du$$

$$u\ln u - u + C = (\ln x)(\ln(\ln x)) - \ln x + C$$

$$\int \frac{\cos^3 x}{\csc x}\,dx \qquad \csc x = \frac{1}{\sin x}$$

$$\int \frac{\cos^3 x}{1/\sin x}\,dx$$

$$\int \cos^3 x \sin x\,dx \qquad u = \cos x \qquad du = -\sin x\,dx$$

$$\int u^3(-du)$$

$$-\frac{1}{4}u^4 + C = -\frac{1}{4}\cos^4 x + C$$

$$\int \frac{x}{x^4+9}\,dx$$

$$x^4+9 = (x^2+\sqrt{6}x+3)(x^2-\sqrt{6}x+3)$$

Instead, $u = x^2$

$$du = 2x\,dx$$

$$\int \frac{1}{u^2+9}\cdot\frac{1}{2}\,du$$

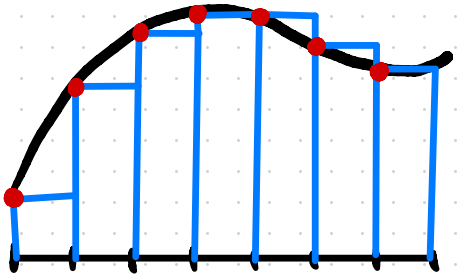
$$\frac{1}{2}\cdot\frac{1}{3}\tan^{-1}\frac{u}{3} + C$$

$$\frac{1}{6}\tan^{-1}\frac{x^2}{3} + C$$

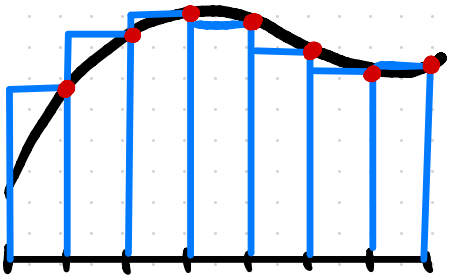
Approximate Integration (7.7)

We have already seen how to use Riemann sums with left endpoints, right endpoints, or midpoints to approximate definite integrals.

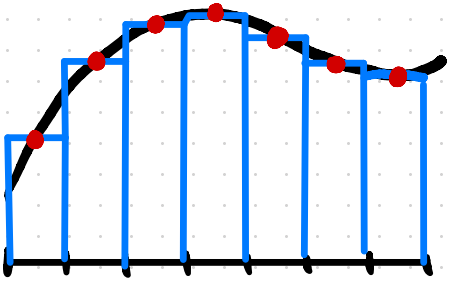
$$\int_a^b f(x)\,dx \qquad n=7 \text{ subintervals}$$



Left endpoints



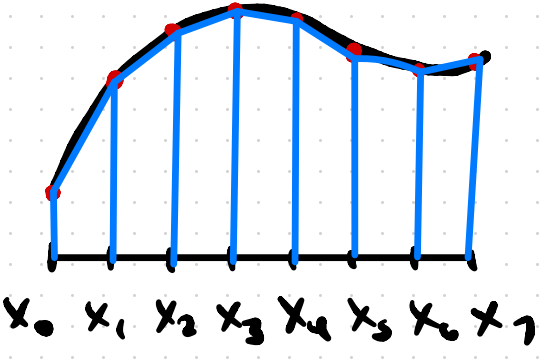
Right endpoints



Midpoints

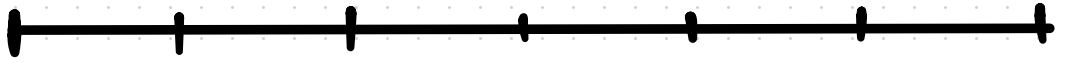
Trapezoidal rule: Subdivide $[a,b]$ into n equal subintervals of length $\Delta x = \frac{b-a}{n}$. Let $[x_{i-1}, x_i]$ be the ith interval. The trapezoidal rule approximation is

$$\int_a^b f(x)\,dx \approx \frac{\Delta x}{2}\Big(f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(x_n) \Big)$$



$\int_0^3 \sqrt{1+x^3}\,dx$ Approximate with 6 equal subintervals.

$\Delta x = 0.5$



$\sqrt{1}$ $\sqrt{1.125}$ $\sqrt{2}$ $\sqrt{4.375}$ $\sqrt{9}$ $\sqrt{16.625}$ $\sqrt{28}$

$$\int_0^3 \sqrt{1+x^3}\,dx \approx \frac{0.5}{2}\left(\sqrt{1} + 2\sqrt{1.125} + 2\sqrt{2} + 2\sqrt{4.375} + 2\sqrt{9} + 2\sqrt{16.625} + \sqrt{28}\right)$$

$$\approx 7.3948$$

$$T_n = \frac{1}{2}(L_n + R_n)$$

trapezoidal

left endpoints

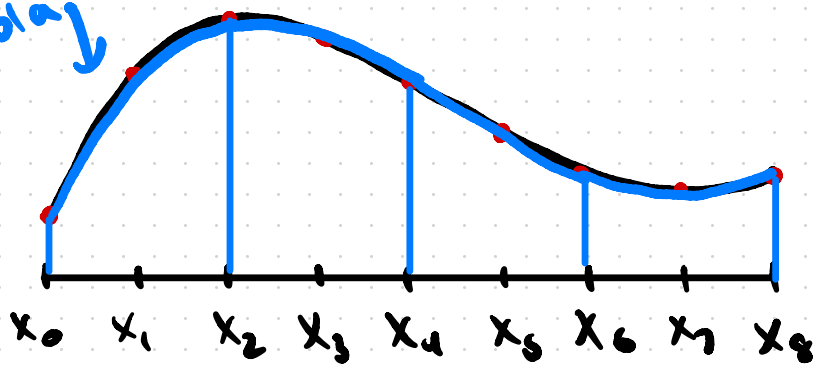
right endpoints

Simpson's rule: Subdivide $[a,b]$ into n equal subintervals of length $\Delta x = \frac{b-a}{n}$. Let $[x_{i-1}, x_i]$ be the ith interval. **Assume n is even.**

Simpson's rule of approximation is

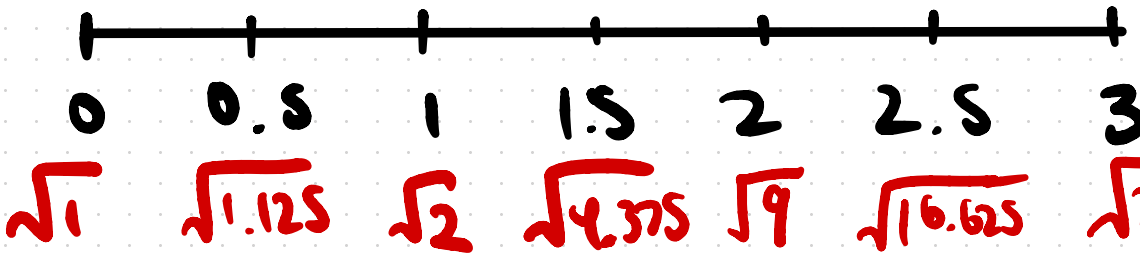
$$\int_a^b f(x)\,dx \approx \frac{\Delta x}{3}\Big(f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \cdots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)\Big)$$

approximate with a parabola



$\int_0^3 \sqrt{1+x^3}\,dx$ Approximate with 6 equal subintervals.

$\Delta x = 0.5$



$$\int_0^3 \sqrt{1+x^3}\,dx \approx \frac{0.5}{3}\left(\sqrt{1} + 4\sqrt{1.125} + 2\sqrt{2} + 4\sqrt{4.375} + 2\sqrt{9} + 4\sqrt{16.625} + \sqrt{28}\right)$$

$$\approx 7.33998$$

$$\int_0^3 \sqrt{1+x^3}\,dx \approx 7.341407\ldots$$

We know that the shown digits are accurate because the approximation methods used have error bounds which guarantee that the true answer is within some range of the approximation. This range decreases rapidly as the number of subintervals increase.

$$S_{2n} = \frac{1}{3}T_n + \frac{2}{3}M_n$$

Simpson's

trapezoidal

midpoint